SAMMY Technologies

UNIVERSAL 3D SIGN LANGUAGE INFRASTRUCTURE | AI-ASSISTED • LEXICON-FIRST | $500K-$1M RAISE

Deterministic 3D Spatial Sign Language for Enterprise Compliance.

Over 70M Deaf individuals globally rely on visual-gestural sign language. Because natural sign is a 3D spatial language with no written orthography, black-box AI video models fail. SAMMY solves bidirectional sign synthesis through a deterministic kinematic compiler running at 60 FPS in browser WebGL—slashing cloud GPU rendering fees and streaming bandwidth by 95%+.

$234M
INITIAL BEACHHEAD TAM (AVIATION + ER)

60 FPS / Δ = 0.000
CLIENT WEBGL • INVARIANT BONE LENGTHS

$500K – $1.0M
PRE-SEED SAFE TARGET ($5M CAP)

LIVE PROTOTYPE STATUS: 34 ASL Signs Audited • 27-DOF Skeletal Hand/Arm Rig • Sub-100ms Latency • Zero Cloud GPUs — INTERACTIVE DEMO ONLINE

Why Legacy Avatars & Big Tech AI Video Fail

FLAW 01 • COMMUNITY

Student "AI Gloves" & Toys

Well-meaning engineering students repeatedly invent sensor gloves identifying 20 isolated signs. The Deaf community fiercely rejects them: natural sign language is conveyed primarily through facial non-manual markers, spatial syntax, and torso posture—not isolated hand flexions.

FLAW 02 • UNIT ECONOMICS

Optical Mocap Avatars

Incumbents pay $100K+ recording human signers in optical mocap suits, locking them to narrow, rigid pre-recorded clips. They cannot scale to open-domain vocabulary, and they burn expensive cloud GPUs streaming heavy MP4 video to screens—running up unsustainable bandwidth bills.

FLAW 03 • HALLUCINATION

Generative Video Diffusion

Big Tech (Google, Meta) treats sign language as a 2D pixel problem. Video diffusion models suffer from catastrophic topological hallucinations: phantom 6th fingers, rubber-band bone stretching, $0.20/query compute fees, and 20-second latencies that make dynamic transit announcements impossible.

The Regulatory Compliance Vacuum:
Under **14 CFR § 382.53(a)(3)** and Title II of the ADA, airlines, airports, and hospitals face statutory liability to provide prompt access to the same information. Yet today, operators have **literally zero dynamic software to buy**.

The Tripartite Architecture: Decoupling Spatial Language

PILLAR 01 • LIVE PROTOTYPE — 27-DOF RIG

1. 3D Kinematic Engine

A deterministic active skeletal rig that compiles text and phonological lexicons (HamNoSys, ASL-LEX 2.0) into exact, repeatable 3D joint movements with invariant bone lengths (Δ = 0.000).

- Live interactive WebGL demo
- Zero mocap studio recording

PILLAR 02 • MILESTONE 2 — FACIAL AUs

2. Neural Humanization

Layers realistic micro-dynamics, facial Action Units (eyebrow raises, mouth morphemes), and emotion over the physical skeleton. Directly crosses the Uncanny Valley to eliminate robotic avatar stiffness.

- Non-manual grammar compliance
- Fluid co-articulation blending

PILLAR 03 • MILESTONE 3 — VISION CHANNEL

3. Video Reading (Perception)

Reads consumer camera video to track 3D hands and facial markers in real time. Translates continuous visual signing back into spoken text and audio, delivering true bidirectional communication.

- Sign → Text reverse translation
- Two-way passenger/gate dialog

COMMUNITY GOVERNANCE

Hearing Founder Humility & Deaf Leadership:
We allocate **$100,000 of our Pre-Seed round** to pay certified native Deaf linguistic researchers and recruit a Deaf co-designer into core executive leadership before commercial deployment.

$100K
DEAF CO-DESIGN ALLOCATION

ARCHITECTURAL MOAT • SCALABILITY

Unit Economics: Kilobytes of JSON vs. Megabytes of Video

COST REDUCTION
95%+ INFRASTRUCTURE SAVINGS

TECHNICAL VECTOR	LEGACY MOCAP AVATARS	GENERATIVE VIDEO DIFFUSION	SAMMY KINEMATIC COMPILER
Studio Recording	Requires $100K+ optical mocap studios	Requires massive pre-training video sets	**Zero mocap recording required (Procedural)**
Network Bandwidth	5 – 20 MB per video stream	High-bandwidth MP4 streaming	**~5 KB lightweight JSON tokens (95%+ savings)**
Cloud Compute Fees	Cloud video servers ($50–$100/mo/screen)	$0.15–$0.30 per query (A100 clusters)	**$0.00 cloud rendering fees (Client WebGL)**
Execution Hardware	Dedicated media playback boxes	Heavy server GPU infrastructure	**Runs locally in browser on existing gate displays**
Latency	1 – 3 second video buffering	15 – 30 second inference latency	**Sub-100ms real-time procedural compilation**
Topological Rigor	Rigid, robotic pre-recorded clips	Frequent finger hallucination & bone stretch	**Deterministic kinematics (Δ = 0.000 variance)**

~5 KB
PAYLOAD PER GATE ANNOUNCEMENT

60 FPS
CLIENT-SIDE WEBGL RENDERING

>90%
TARGET GROSS SOFTWARE MARGIN

MARKET SIZING • ENTERPRISE B2B SAAS

$234M Initial Compliance Beachhead

BOTTOM-UP TAM
NON-DISCRETIONARY COMPLIANCE

BEACHHEAD 01 • AVIATION

$144M ARR

Commercial Airline Boarding Gates

15,000 domestic commercial boarding gates across primary hub airports. Carriers and airport authorities share statutory responsibility under 14 CFR § 382.53(a)(3) for timely accessible gate information.

GATE TARGETS
15,000 Gates

PRICING
$800 / gate / mo

BEACHHEAD 02 • HEALTHCARE

$90M ARR

Hospital Emergency Department Triage

5,000 acute hospital emergency intake desks subject to Title III of the ADA and Section 1557 of the Affordable Care Act, requiring immediate communication for critical intake.

TRIAGE DESKS
5,000 Desks

PRICING
$1,500 / desk / mo

Closing the Disability Wealth Gap:
SAMMY's consumer mobile application remains **permanently free for Deaf travelers**. Transit authorities, airlines, and hospital systems pay enterprise subscriptions out of compliance budgets.

$2.34M
1% BEACHHEAD CAPTURE ARR

Why Back SAMMY at Pre-Seed: The 4-in-1 Asymmetry

The downside risk is bounded to technical milestones; the upside return is asymmetric: if this works, SAMMY becomes the only complete framework uniting natural sign language translation with spatial AI.

01 • PROTOCOL MOAT

The Stripe for Sign Language

Big Tech burns capital on generative video models that hallucinate fingers. SAMMY builds the deterministic kinematic compiler, owning the underlying SDK standard for spatial headsets, kiosks, and operating systems globally.

02 • REGULATORY LOCK-IN

DOJ / DOT Mandates

Airports face active compliance exposure under 14 CFR § 382.53 for acoustic gate changes, but have zero dynamic software to buy. Being first to gate compliance creates an unshakeable B2B enterprise beachhead.

03 • PORTFOLIO SYNERGY

The Evolution of Ava

Enable backed Ava for speech-to-text captions. English captions only serve people who read text; SAMMY completes the stack for native 3D spatial signers, creating an unassailable accessibility portfolio.

04 • VALUATION FLOOR

Pre-Seed Entry Point

Backing us at our **$5M pre-seed cap** secures ground-floor ownership of the foundational standard before our 500-phrase lexicon and live commercial airport trial re-rates the company to a $20M+ Series A valuation.

FINANCING • CAPITAL ALLOCATION

Pre-Seed Raise: $500K – $1.0M SAFE ($5M Cap)

RUNWAY: 18–24 MONTHS
VALUATION CAP: $5.0M POST-MONEY

Use of Funds Breakdown ($1.0M Target)

Core Engineering & Founder Runway:	$350,000
Deaf Co-Design & Linguistic Leadership:	$100,000
3D Assets, Rigging & Facial Pipeline:	$150,000
Airport Pilot & Edge Hardware:	$150,000
IP, Legal (DOT/FAA) & Reserve Buffer:	$250,000

Note: $500K minimum tranche fully de-risks the linguistic and neural milestones; $1.0M accelerates the live commercial airport gate deployment.

3 Critical De-Risking Milestones

- **Milestone 01: 500-Phrase Deaf-Audited Airport Lexicon**
  Expand beyond current 34 signs to complete 500 validated gate, delay, and emergency aviation phrases audited and certified by native Deaf linguists.
- **Milestone 02: Facial Action Units & Neural Humanization**
  Integrate non-manual grammar (eyebrow, mouth morphemes) to conquer the Uncanny Valley and achieve native linguistic acceptance.
- **Milestone 03: Live Commercial Airport Gate Trial**
  Deploy SAMMY on active boarding gate screens in partnership with an initial US regional/hub airport authority under 14 CFR § 382.53.

Team, Traction & 15-Minute Video Call

FOUNDER & LEAD TECHNOLOGIST

Iniyan Andrews Joseph

Deep-tech researcher specializing in inverse kinematics, computational linguistics, and spatial AI. Built the initial 27-DOF WebGL kinematic engine, eliminating the need for optical mocap suits or cloud GPUs.

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LIVE 45-SECOND PROTOTYPE

Try the 3D Kinematic Engine

Experience the real-time procedural rig running at 60 FPS directly in your browser with zero plugins or installation required:

`https://joeytribb.github.io/sign-language-kinematics/?v=5`

Features 34 validated signs, active finger kinematics, and real-time bone invariance verification.

Proposed 15-Minute Live Video Demonstration:
Because sign language is an embodied 3D spatial language, a video call is the ideal medium to demonstrate the kinematics and discuss roadmap alignment live.

Tuesday, Oct 27 @ 10:00 AM ET

Thursday, Oct 29 @ 2:00 PM ET